MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special

Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Value Addition Course (VAC) - MG3VACPES201 -LIFESTYLE DISEASES & PHYSICAL ACTIVITY

(2024 ADMISSION ONWARDS)

Answer Key

Part A MCQ

Answer all questions each question carries 1 marks

1. B) Hypertension

2. B) Walking and climbing stairs

3. A) Protein

4. B) Whole grains, fruits, vegetables, lean proteins

5. C) Cycling

Part B SHORT ANSWER QUESTIONS
Answer any 5 questions Each question carries 2 marks

6. Define health and explain its dimensions. [U] / [CO1]

- **Health** is a state of complete physical, mental and social well-being, and not merely the absence of disease or infirmity. **(1 mark)**
- Major dimensions of health include **physical, mental, social, emotional, spiritual and environmental health**. Any two/appropriate dimensions with explanation may be accepted. **(1 mark)**

7. Give two economic consequences of an unhealthy lifestyle. [An] / [CO2]
Any **two**:

- Increased medical and healthcare expenses.
- Loss of income due to illness and absenteeism.
- Reduced work productivity.
- Increased burden on families and society.
- Premature disability or early retirement leading to loss of earning capacity.
  (1 mark each × 2 = 2 marks)

8. What is obesity and why is it considered a lifestyle disease? [U] / [CO2]

- **Obesity** is a condition in which excessive body fat accumulates to a level that may impair health; it is commonly assessed using **BMI (30 kg/m² or above in adults)**. **(1 mark)**
- It is considered a **lifestyle disease** because it is strongly associated with unhealthy dietary habits, excessive calorie intake, physical inactivity and sedentary behaviour. **(1 mark)**

9. How can a lack of physical activity affect heart health? [U] / [CO3]

- Physical inactivity can contribute to **obesity, high blood pressure and unhealthy cholesterol levels**. **(1 mark)**
- These factors increase the risk of **heart disease, coronary artery disease and stroke**. **(1 mark)**

10. Give four sources of carbohydrates. [U] / [CO6]
Any **four**:

- Rice
- Wheat
- Bread
- Potatoes
- Maize/Corn
- Fruits
- Sugar
- Cereals
- Pulses/legumes
  (0.5 mark each × 4 = 2 marks)

11. Define functional training. [K] / [CO5]

- **Functional training** is a form of exercise that uses movements and exercises designed to improve the ability to perform **everyday activities or sport-specific movements efficiently and safely**. **(2 marks)**

12. Write one difference between lifestyle diseases and infectious diseases. [U] / [CO1]

Lifestyle Diseases	Infectious Diseases
Mainly associated with unhealthy lifestyle factors such as poor diet, physical inactivity and smoking, and are generally not transmitted from person to person.	Caused by pathogens such as bacteria, viruses, fungi or parasites and can be transmitted directly or indirectly.

Answer Key – Part C: Short Essay Questions

Answer any 4 questions. Each question carries 5 marks.
[5 × 4 = 20 Marks]

13. A male college student weighs 72 kg and is 1.68 m tall. Find the student's BMI and classify the weight status. [A] / [CO6]

Formula:

BMI=Height2 (m)Body Weight (kg)

Calculation:

BMI=(1.68)272 BMI=2.822472=25.51 kg/m2

BMI = 25.5 kg/m²

Classification (WHO adult classification):

- Below 18.5 – Underweight
- 18.5–24.9 – Normal weight
- **25.0–29.9 – Overweight**
- 30.0 and above – Obesity

Answer: The student's BMI is **25.5 kg/m²**, which falls under the **overweight** category.

Marking scheme (5 marks):

- Correct formula – 1 mark
- Correct substitution – 1 mark
- Correct calculation – 1 mark
- Correct BMI value – 1 mark
- Correct classification – 1 mark

14. Explain how a balanced diet is linked to healthy living. [An] / [CO2]

A **balanced diet** provides all essential nutrients—carbohydrates, proteins, fats, vitamins, minerals, fibre and water—in appropriate amounts.

Its role in healthy living includes:

1. **Provides energy** required for daily activities and exercise.
2. **Supports growth and repair** of body tissues through adequate protein and nutrients.
3. **Maintains healthy body weight** by providing appropriate amounts of energy.
4. **Strengthens immunity** and helps the body resist diseases.
5. **Prevents nutritional deficiencies and lifestyle diseases** such as obesity, type 2 diabetes and cardiovascular diseases.
6. **Improves mental and physical performance** and promotes overall well-being.

Marking scheme: Any five relevant points – **1 mark each**.

15. Which laboratory tests are commonly performed to assess obesity-related complications? [U] / [CO2]

Common laboratory investigations include:

1. **Fasting blood glucose / HbA1c** – to assess diabetes and blood glucose control.
2. **Lipid profile** – to measure total cholesterol, LDL, HDL and triglycerides.
3. **Liver function tests (LFT)** – to identify possible fatty liver and other liver abnormalities.
4. **Kidney function tests** – such as serum creatinine and blood urea to assess kidney health.
5. **Thyroid function tests** – particularly TSH, when thyroid dysfunction is suspected.
6. **Other tests when clinically indicated** – such as fasting insulin or uric acid.

Marking scheme (5 marks):

- Any five appropriate tests with their purpose – **1 mark each**.

16. How does exercise help in the prevention of back pain? [U] / [CO3]

Regular exercise can help prevent back pain through the following mechanisms:

1. **Strengthens core muscles**, including the abdominal and back muscles, which support the spine.

2. **Improves flexibility** of the spine, hips and surrounding muscles.
3. **Improves posture**, reducing unnecessary stress on the back.
4. **Improves muscular endurance and body mechanics**, helping individuals perform daily activities safely.
5. **Helps maintain healthy body weight**, reducing excessive load on the spine.
6. **Improves blood circulation and mobility**, which supports healthy muscles and joints.

Examples include **walking, swimming, core-strengthening exercises, stretching and appropriate resistance exercises**.

Marking scheme: Any five relevant points – **1 mark each**.

17. Differentiate between dumbbell and kettlebell training. [U] / [CO5]

Dumbbell Training	Kettlebell Training
Consists of weights with a handle between two equal-weight ends.	Consists of a ball-shaped weight with a single top handle.
Weight is generally held close to the body.	The weight can be held away from the body and moved dynamically.
Commonly used for controlled strength and isolation exercises.	Commonly used for dynamic, full-body movements.
Examples: biceps curl, shoulder press and dumbbell row.	Examples: kettlebell swing, Turkish get-up and goblet squat.
Useful for unilateral and bilateral strength training.	Particularly useful for developing strength, power, coordination and cardiovascular conditioning.

Marking scheme:

- Any **five valid differences – 1 mark each**.

18. Explain the connection between screen time and anxiety. [E] / [CO6]

Excessive screen time, particularly prolonged social media and entertainment use, may be associated with increased anxiety through several pathways:

1. **Reduced physical activity:** Excessive screen use may replace exercise and active recreation, which can negatively affect mental well-being.
2. **Sleep disturbance:** Late-night screen use can reduce sleep duration and quality, contributing to stress and anxiety.

3. **Social comparison:** Social media may encourage comparison with idealized images and lifestyles, leading to feelings of inadequacy or worry.
4. **Information overload:** Continuous exposure to notifications, news and online content can increase mental stress.
5. **Reduced face-to-face interaction:** Excessive screen use may reduce meaningful social interaction and support.
6. **Poor concentration and mental fatigue:** Constant switching between apps and notifications may increase psychological strain.

Conclusion: Moderate and purposeful screen use, adequate sleep, regular physical activity and healthy face-to-face social interaction can help support better mental well-being.

Part D – Essay Questions: Suitable Answer Key

Answer any ONE question. Each question carries 15 marks.
[15 × 1 = 15 Marks]

19. Discuss in detail the role of nutrition in preventing chronic diseases. [An] / [CO6]

Introduction – 2 marks
Nutrition plays a major role in maintaining health, supporting normal body functions and reducing the risk of chronic diseases. A balanced diet containing appropriate amounts of carbohydrates, proteins, healthy fats, vitamins, minerals, fibre and water helps prevent many lifestyle-related diseases.

Role of nutrition in prevention – 10 marks

1. **Prevention of obesity – 2 marks**
    - Balanced energy intake helps maintain healthy body weight.
    - Limiting excess calories, sugary foods and highly processed foods reduces the risk of obesity.
2. **Prevention of cardiovascular diseases – 2 marks**
    - Adequate intake of fruits, vegetables, whole grains and fibre supports heart health.
    - Limiting saturated fat, trans fat, excess salt and added sugars helps control cholesterol and blood pressure.
3. **Prevention and management of diabetes – 2 marks**
    - High-fibre foods and appropriate carbohydrate intake help regulate blood glucose.
    - Reducing refined carbohydrates and excessive added sugar can reduce the risk of type 2 diabetes.
4. **Prevention of hypertension – 1 mark**

    - Reducing excess sodium/salt intake and consuming potassium-rich foods can help maintain healthy blood pressure.
5. **Prevention of certain cancers – 1 mark**
    - A diet rich in fruits, vegetables and whole grains provides fibre and protective nutrients and is associated with a lower risk of some cancers.
6. **Healthy bones and muscles – 1 mark**
    - Adequate calcium, vitamin D, protein and other nutrients support bone and muscle health and reduce the risk of deficiencies.
7. **Healthy digestive system – 1 mark**
    - Dietary fibre and adequate water support bowel function and help prevent constipation and other digestive problems.

Healthy dietary practices – 2 marks

- Eat a variety of fruits and vegetables.
- Prefer whole grains and high-fibre foods.
- Choose lean protein and healthy fats.
- Limit salt, added sugars, saturated fats and highly processed foods.
- Maintain adequate hydration and appropriate portion sizes.

Conclusion – 1 mark
A balanced and nutrient-rich diet, combined with regular physical activity and other healthy lifestyle practices, is an important strategy for preventing and controlling chronic diseases.

Total: 15 marks

20. What are psychosomatic disorders, and how do stress, anxiety, and strong emotions contribute to their development? [U] / [CO4]

Introduction – 2 marks
Psychosomatic disorders are physical symptoms or health problems that are influenced or worsened by psychological factors such as stress, anxiety and emotional disturbances. The symptoms are real, although psychological factors can play an important role in their development or severity.

Meaning of psychosomatic disorders – 3 marks

- **Psycho** refers to psychological or mental processes.
- **Somatic** refers to physical or bodily symptoms.
- Examples may include **tension headaches, certain gastrointestinal problems, muscle tension and stress-related pain**.
- Psychological stress does not mean that the symptoms are imaginary; mind and body interact continuously.

Role of stress – 3 marks

1. Stress activates the body's **fight-or-flight response**.
2. Increased sympathetic nervous system activity and stress hormones can affect heart rate, muscle tension, digestion and sleep.
3. Persistent stress may contribute to headaches, gastrointestinal discomfort, fatigue and other physical symptoms.

Role of anxiety – 2 marks

1. Anxiety produces continuous worry and physiological arousal.
2. It may cause muscle tension, rapid heartbeat, sweating, sleep problems and digestive disturbances.

Role of strong emotions – 3 marks

1. Anger, fear, grief and emotional conflict can produce physiological changes.
2. Strong emotions may increase muscle tension and alter heart rate, breathing and digestion.
3. When emotional stress is prolonged, physical symptoms may become frequent or severe.

Prevention/management and conclusion – 2 marks

- Stress management, regular physical activity, adequate sleep, relaxation techniques, social support and professional psychological/medical care when necessary can help.
- **Conclusion:** Psychological well-being and physical health are closely connected, and managing stress and emotions is important for overall health.

Total: 15 marks

21. Evaluate the role of physical activity in reducing the burden of lifestyle-related diseases in society. [An] / [CO3]

Introduction – 2 marks
Lifestyle-related diseases are chronic conditions associated with factors such as physical inactivity, unhealthy diet, obesity, smoking and prolonged sedentary behaviour. Regular physical activity is an important low-cost strategy for preventing and managing these diseases.

Role of physical activity – 8 marks

1. **Prevents obesity – 1.5 marks**
   Regular activity increases energy expenditure and helps maintain healthy body weight and body composition.
2. **Reduces cardiovascular disease risk – 1.5 marks**
   Exercise improves cardiovascular fitness and can help control blood pressure, blood lipids and other cardiovascular risk factors.
3. **Helps prevent type 2 diabetes – 1.5 marks**
   Physical activity improves insulin sensitivity and glucose utilization, thereby helping regulate blood glucose.
4. **Controls hypertension – 1 mark**
   Regular aerobic and resistance exercise can contribute to healthy blood pressure levels.
5. **Improves musculoskeletal health – 1 mark**
   Strength and weight-bearing activities improve muscle strength, bone health, balance and functional capacity.
6. **Improves mental health – 1 mark**
   Regular physical activity can reduce stress and symptoms of anxiety and depression and improve mood and well-being.
7. **Supports healthy ageing – 0.5 mark**
   It helps maintain independence, mobility and functional capacity in older adults.

Societal benefits – 3 marks

- Reduces healthcare expenditure associated with preventable chronic diseases.
- Improves productivity and reduces sickness-related absenteeism.
- Promotes healthier and more active communities.
- Reduces the long-term burden on families and healthcare systems.
- Encourages healthy habits among children and young people.

Evaluation/limitations – 1 mark
Physical activity is highly beneficial, but it should be combined with **healthy nutrition, adequate sleep, avoidance of tobacco, responsible alcohol use and appropriate medical care**. Exercise alone cannot eliminate all risk factors for chronic diseases.

Conclusion – 1 mark
Promoting regular physical activity at individual, institutional and community levels can significantly reduce the burden of lifestyle-related diseases and improve the overall health and quality of life of society.

Total: 15 marks